

CodeGraph — Architecture Overview

Git blame tells you who changed it. CodeGraph tells you why.

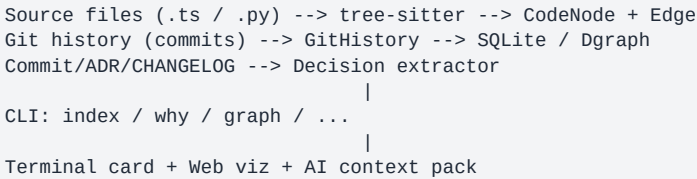
Problem

Developers constantly ask “why is this code written this way?” git blame shows authorship. GitHub search shows text matches. ADR tools require manual writing. AI coding assistants see only the current file. No existing tool connects a line of code to the design decisions that shaped it.

Solution — three layers, one graph

- 1. **Structure** — tree-sitter parses TypeScript/Python into AST nodes and edges (uses / defined_by / contains)
- 2. **Time** — every node carries valid_from / valid_to from real Git history
- 3. **Decisions** — extracted from commit messages, ADRs, and CHANGELOG, linked to the code lines they justify

Architecture



Data model

CodeNode — uid, kind, name, file_path, line_start, line_end, language, valid_from, valid_to, parent_uid

Edge — from_uid, to_uid, kind (uses / defined_by / contains)

Decision — content, reason, alternatives, status, source, source_ref, timestamp, author, constraints, confidence

Seven commands

Command	Purpose
codegraph index <path>	Parse + ingest Git history + store
codegraph why <file>:<line>	Decision card for that line
codegraph graph --at <date>	Code graph at a past moment
codegraph decisions --timeline	Decision evolution chain
codegraph conflicts	Detect contradictory decisions
codegraph export --for-ai	Export AI agent context
codegraph serve	Local web visualization

Tech choices

Layer	Choice	Reason
Parsing	tree-sitter	Multi-language, incremental
Git	GitPython	Read-only real Git history
Graph DB	Dgraph	Multi-hop decision tracing
CLI	Typer + Rich	Fast CLI, readable output
Models	pydantic	Typed data models
Storage	SQLite	Zero-config local default

Status (as of submission)

Status	Item
Done	Product site · CLI (6 commands) · SQLite · real Git history
Done	Decision extraction · time-travel graph --at · 48 tests
Done	tree-sitter TS/Py/Java/Go/Rust/C++ · Web · VSCode · LLM hook
In progress	Dgraph backend (optional live cluster)

